



A Comprehensive Dental Clinic Management System

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What is a Dental Clinic Management System?

A dental clinic management system is a digital platform that helps clinics manage:

- Patients
- Appointments
- Treatments
- Payments
- Communication

It replaces paper records and manual coordination with a centralized system.

Why Dentify?

- **Growing patient load**

Dental clinics are facing increasing numbers of patients, making manual management inefficient.

- **Fragmented systems**

Many clinics use separate tools for appointments, billing, and records, leading to errors and delays.

- **Poor patient experience**

Patients rely on phone calls and physical visits for simple tasks like booking or follow-up.

- **Lack of intelligent automation**

Most existing systems do not use AI to reduce staff workload or improve accessibility.

Need for a Smart & Integrated Solution

There is a strong need for:

- A single unified platform
- Web and mobile access
- Secure medical data handling
- Intelligent assistance
- Real-time communication

Dentify was created to fill this gap.

What is Dentify?

- **All-in-one clinic platform**

A complete system covering administration, clinical work, and patient interaction.

- **Web & mobile based**

Accessible anytime through browsers and smartphones.

- **AI-powered**

Uses artificial intelligence to assist patients and clinics.

- **Role-based**

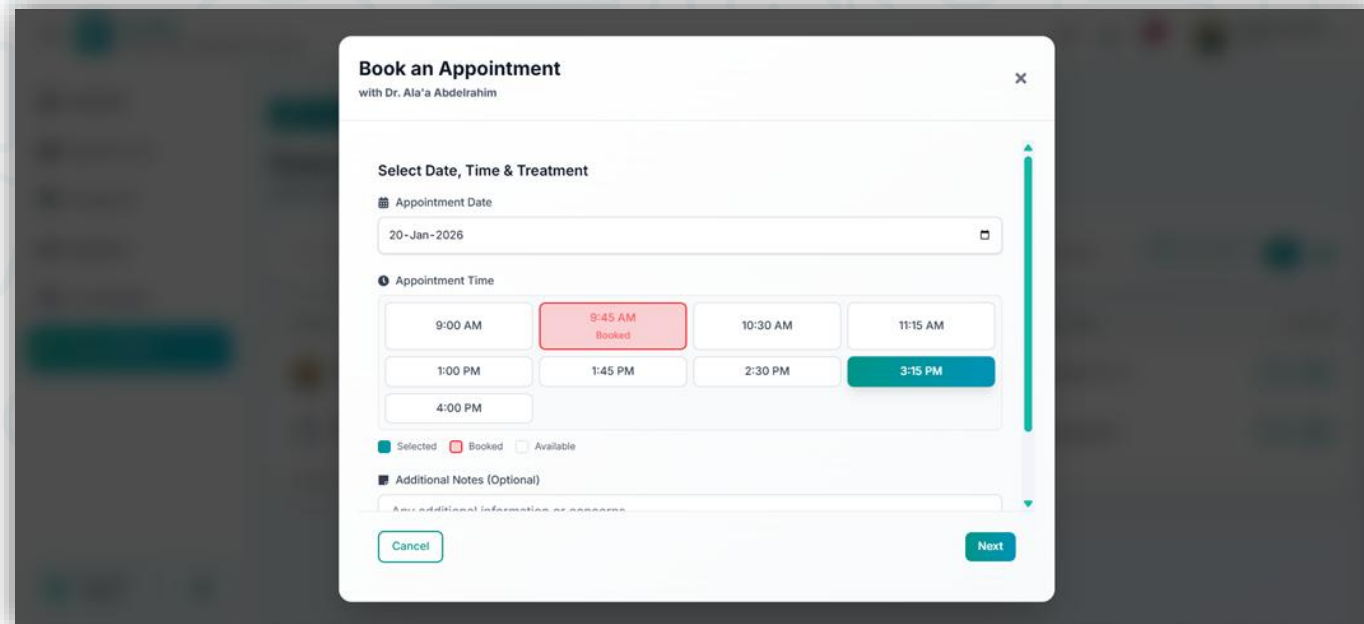
Different interfaces for admins, dentists, patients, and staff.

The background is a light gray pattern of various dental-related icons. These include teeth, toothbrushes, dental chairs, X-rays, dental tools like pliers and probes, and bottles of mouthwash or toothpaste. The icons are scattered across the entire background.

Key Features

Smart Appointment Booking

For real-time scheduling with conflict prevention.



The desktop view of the 'Book an Appointment' form is displayed within a white modal window. The title 'Book an Appointment' is at the top, followed by the text 'with Dr. Ala'a Abdelrahim'. Below this is a section titled 'Select Date, Time & Treatment'. It contains an 'Appointment Date' field with the value '20-Jan-2026' and a calendar icon. Underneath is the 'Appointment Time' section, which shows a grid of time slots. The slots are color-coded: green for 'Selected', red for 'Booked', and white for 'Available'. The 3:15 PM slot is selected. A legend at the bottom of the time grid shows a green square for 'Selected', a red square for 'Booked', and a white square for 'Available'. Below the time grid is an 'Additional Notes (Optional)' section with a text input field and a 'Cancel' button. A 'Next' button is located at the bottom right of the modal.

Book an Appointment
with Dr. Ala'a Abdelrahim

Select Date, Time & Treatment

Appointment Date
20-Jan-2026

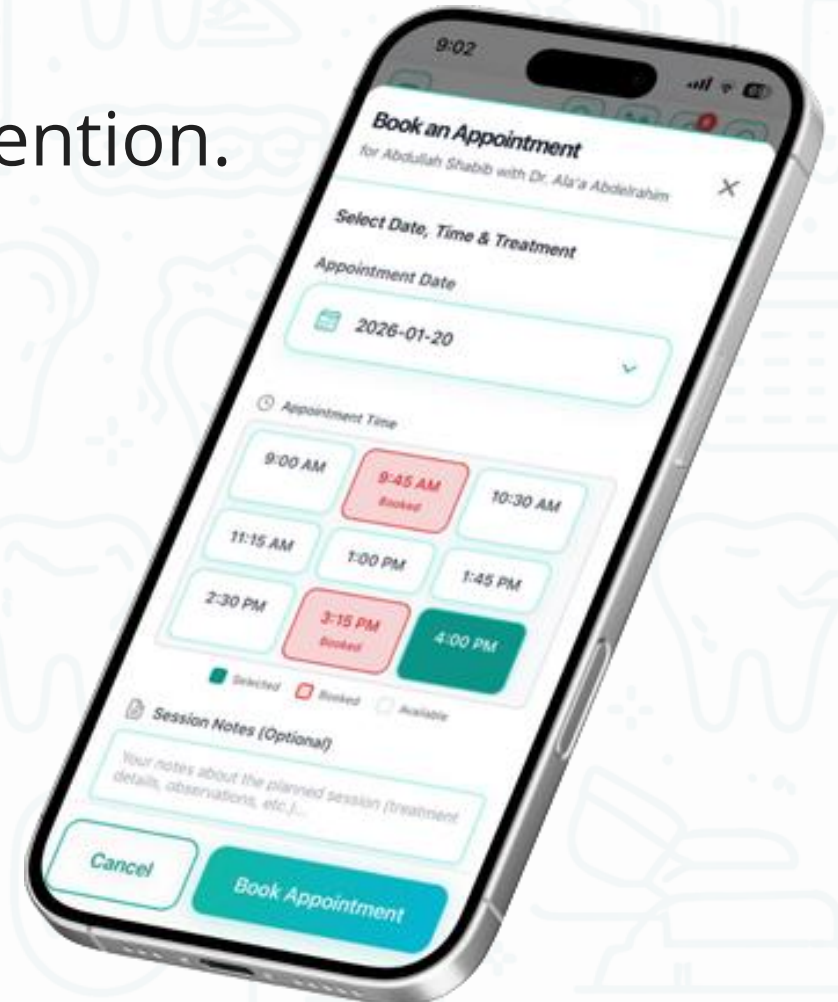
Appointment Time

9:00 AM	9:45 AM Booked	10:30 AM	11:15 AM
1:00 PM	1:45 PM	2:30 PM	3:15 PM Selected
4:00 PM			

Selected Booked Available

Additional Notes (Optional)
Any additional information or concerns

Cancel Next



The mobile view of the 'Book an Appointment' form is shown on a smartphone screen. The title 'Book an Appointment' is at the top, followed by the text 'for Abdullah Shabb with Dr. Ala'a Abdelrahim'. Below this is a section titled 'Select Date, Time & Treatment'. It contains an 'Appointment Date' field with the value '2026-01-20' and a calendar icon. Underneath is the 'Appointment Time' section, which shows a grid of time slots. The slots are color-coded: green for 'Selected', red for 'Booked', and white for 'Available'. The 3:15 PM slot is selected. A legend at the bottom of the time grid shows a green square for 'Selected', a red square for 'Booked', and a white square for 'Available'. Below the time grid is a 'Session Notes (Optional)' section with a text input field and a 'Cancel' button. A 'Book Appointment' button is located at the bottom right of the form.

Book an Appointment
for Abdullah Shabb with Dr. Ala'a Abdelrahim

Select Date, Time & Treatment

Appointment Date
2026-01-20

Appointment Time

9:00 AM	9:45 AM Booked	10:30 AM
11:15 AM	1:00 PM	1:45 PM
2:30 PM	3:15 PM Booked	4:00 PM Selected

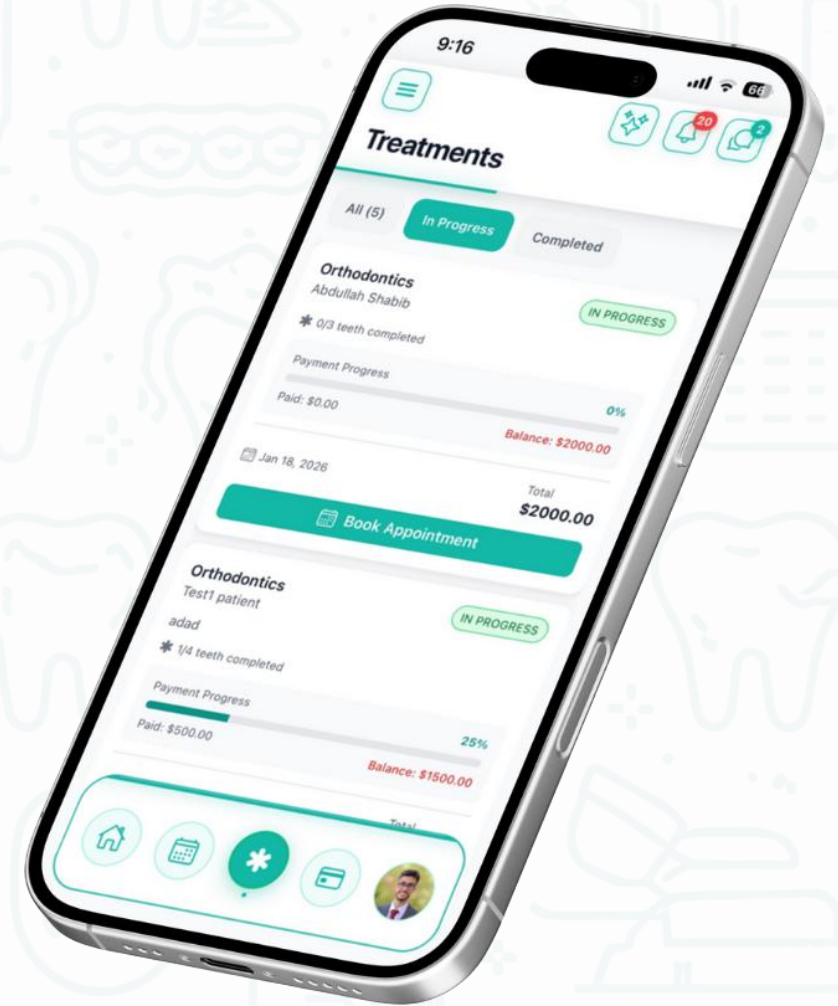
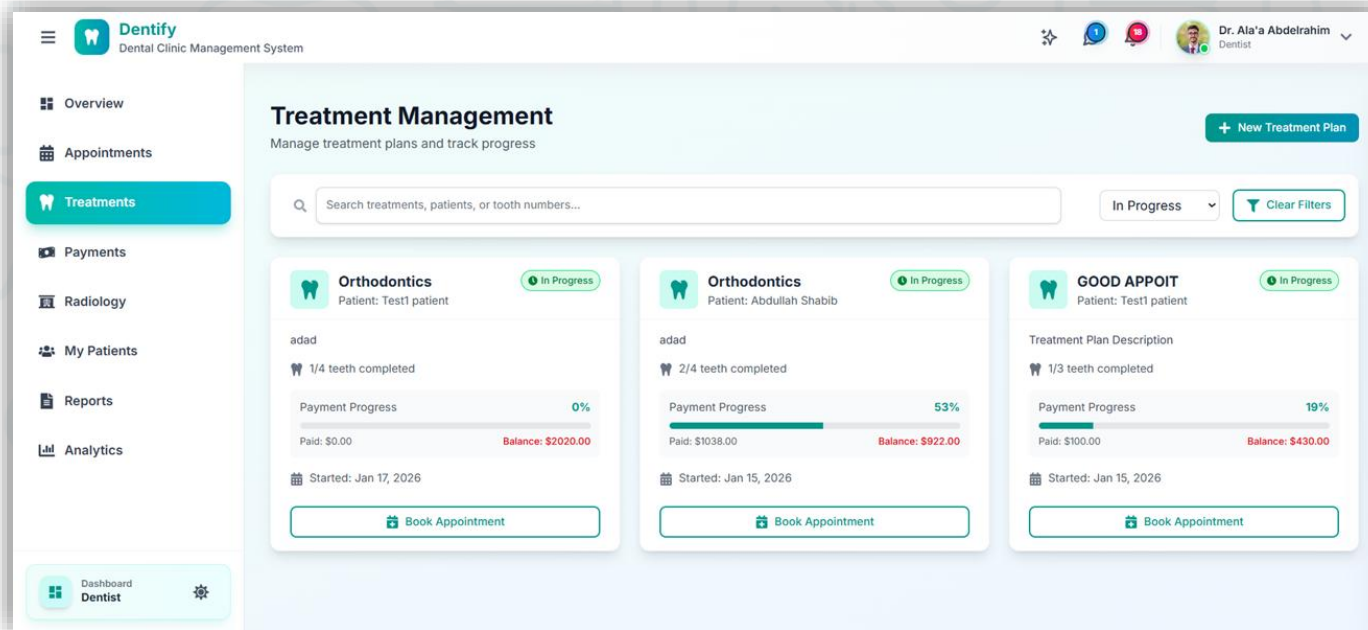
Selected Booked Available

Session Notes (Optional)
Your notes about the planned session (treatment details, observations, etc.)...

Cancel Book Appointment

Patient Record Management

For secure storage of medical history and treatments.



Payment and Billings

For digital payment tracking and receipts.

Add Payment

Total Amount

\$2020.00

Paid Amount

\$0.00

Remaining Balance

\$2020.00

Select Patient *

Test1 patient

Select Treatment *

Orthodontics - Balance: \$2020.00

Payment Amount *

\$

500

Maximum: \$2020.00

Pay Full Balance

Discount (Optional)

\$

20

This discount will reduce the total treatment amount

Cancel

Record Payment

PAYMENT RECEIPT

Receipt Number: #29

CLINIC INFORMATION

PROVIDER & PATIENT INFORMATION

Clinic Name:
Location:
Date of Payment:

Happy Smile Clinic
Rafidia Street
Jan 18, 2026

Dentist:
Abdelrahim
Patient:
Payment Method:

Dr. Ala'a

Test1 patient
Cash

Description

Amount

Orthodontics

\$500.00

Discount Applied

-\$20.00

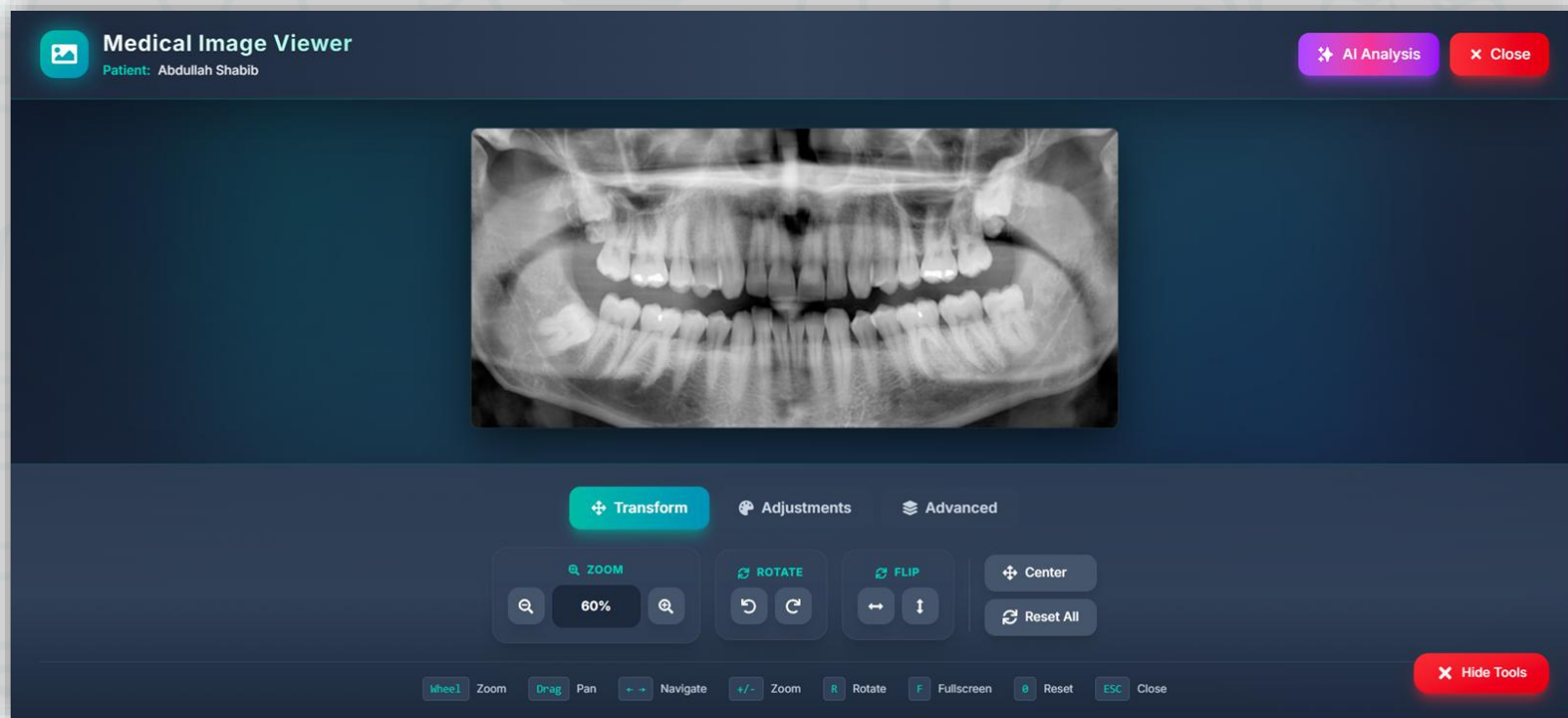
Total Amount Paid: \$500.00

*Includes \$20.00 discount applied to treatment total

This is an official payment receipt. Please retain for your records.
Thank you for your payment.

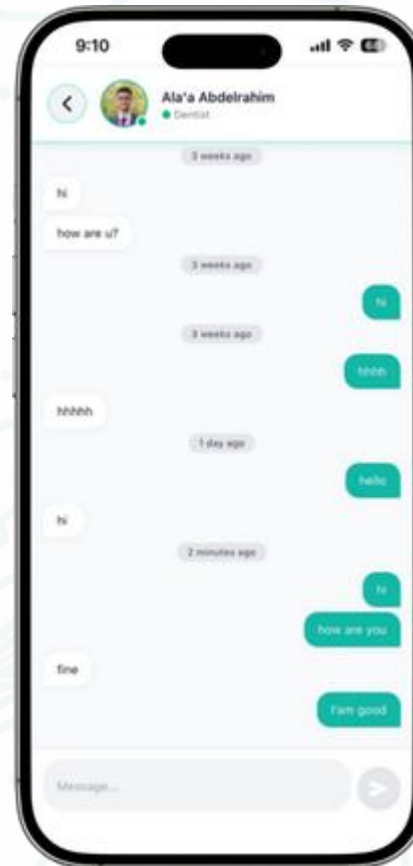
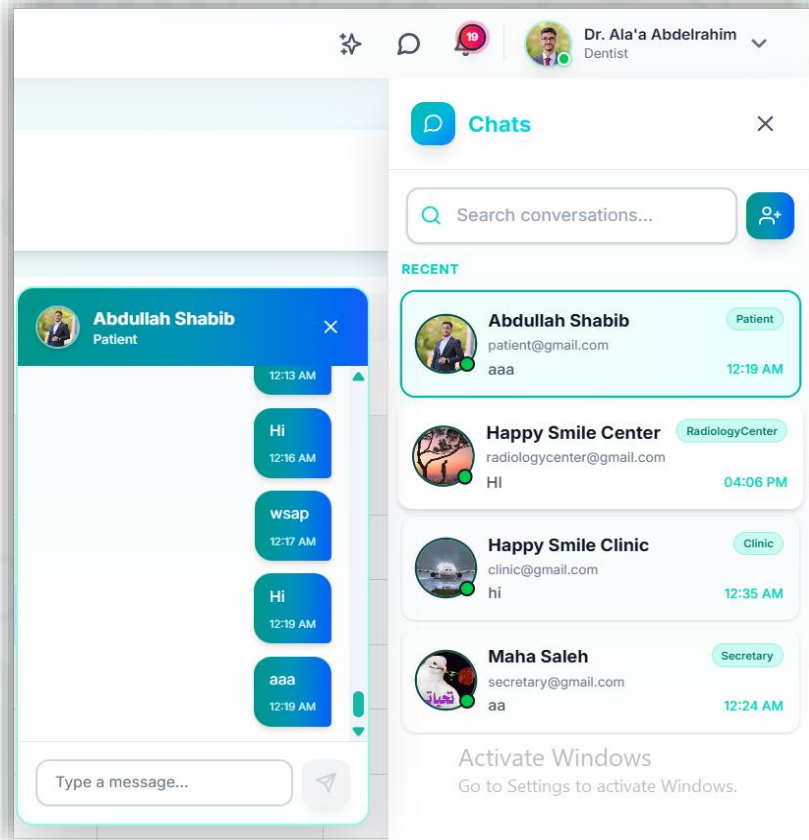
Radiology Integration

For digital imaging requests and results.



Chatting System

For real-time communication between users.

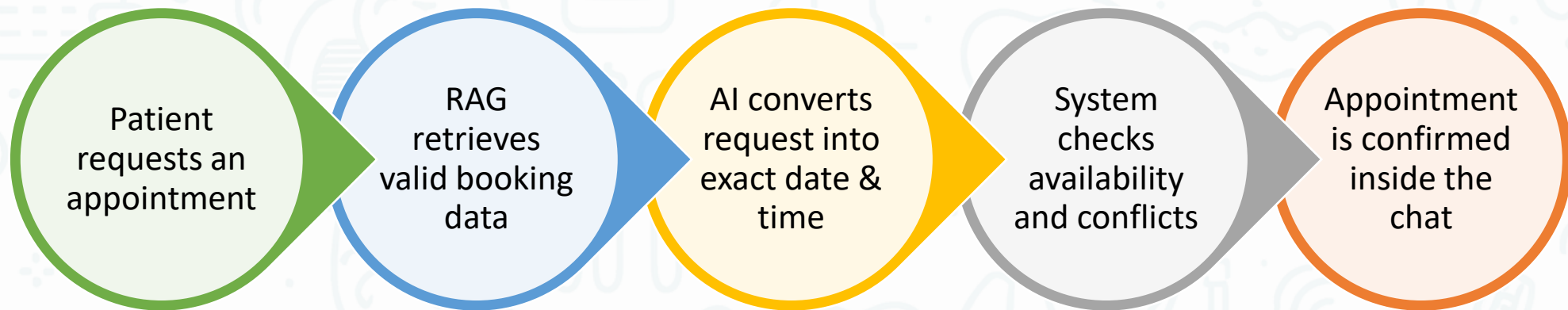


AI Chatbot (1)

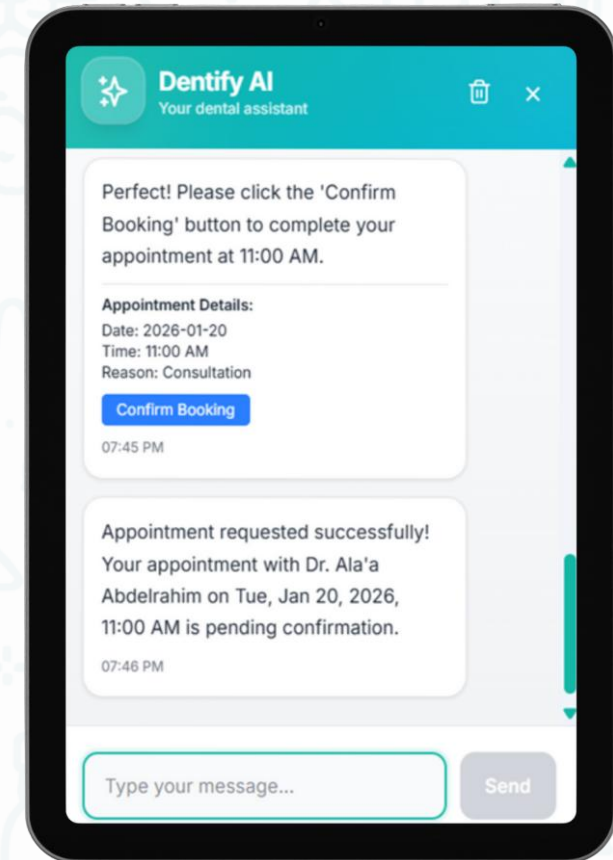
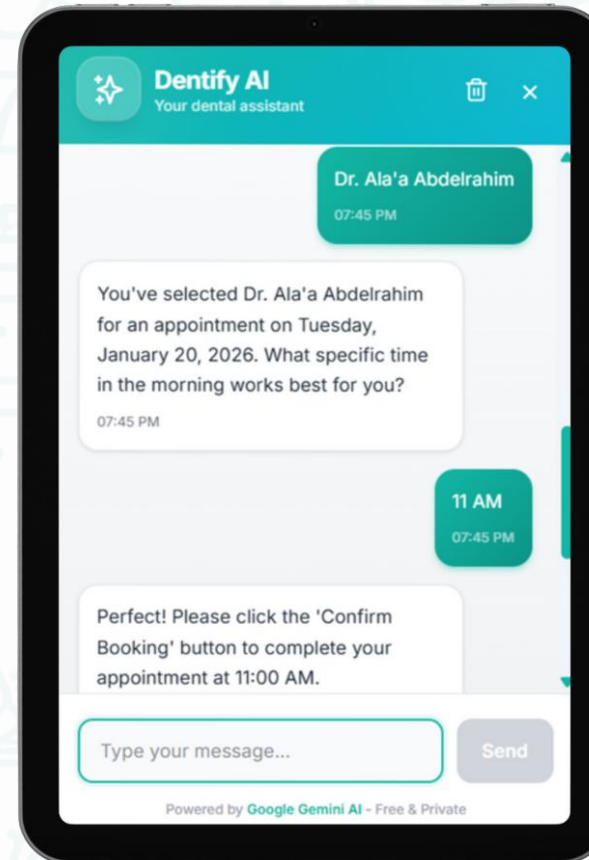
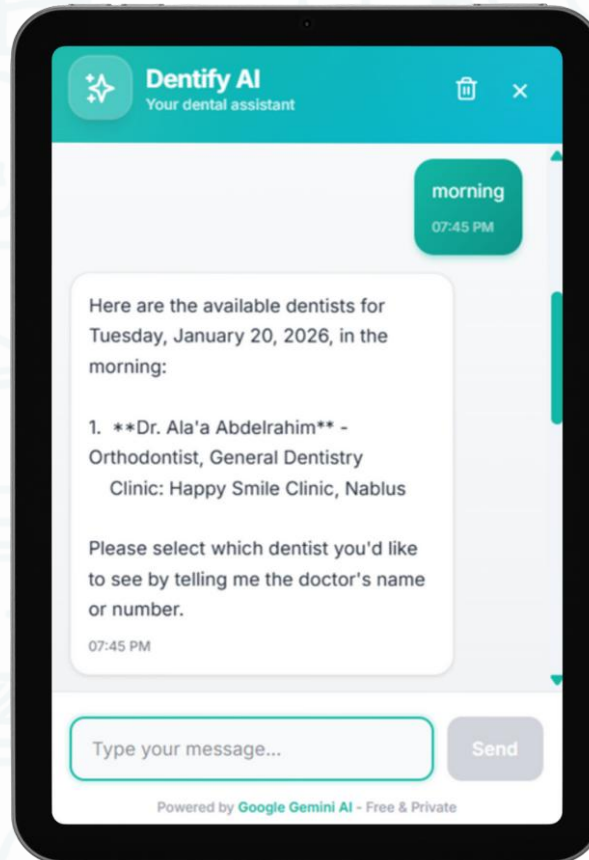
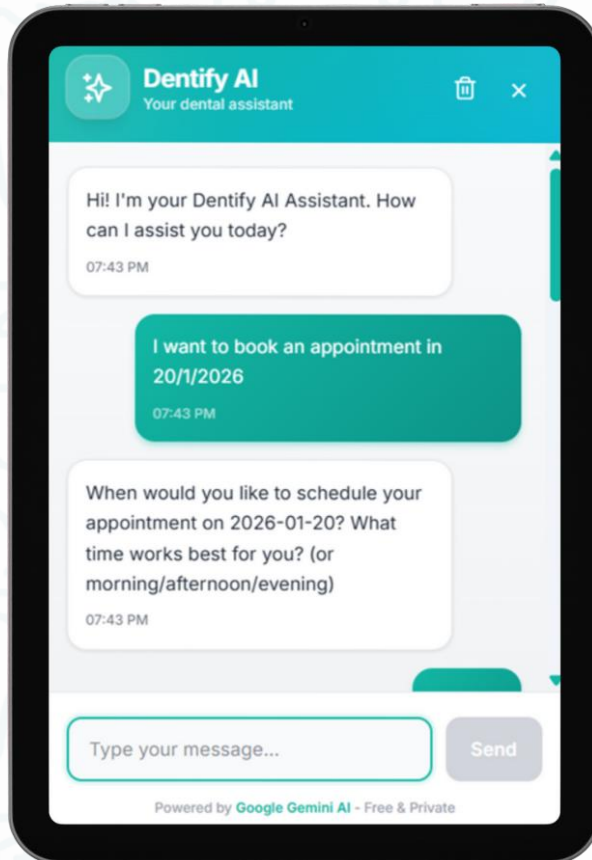
A virtual assistant for booking and inquiries.

- AI chatbot for **booking dental appointments**
- Understands patient messages in natural language
- Built using **Gemini 2.5 Flash**
- **RAG-based booking**
 - ✓ Uses **Retrieval-Augmented Generation (RAG)**
 - ✓ Retrieves real clinic data (dentists, hours, slots)
 - ✓ Ensures accurate and safe bookings

AI Chatbot (2)



AI Chatbot (3)



Software & Technology

Backend



EXPRESS Js



Cloud & Services



Firebase

Frontend



React



React Native



tailwindcss

AI

Gemini



The background is a light gray pattern of various dental-related icons. These include teeth, toothbrushes, dental chairs, X-rays, dental tools like pliers and probes, and bottles of mouthwash or toothpaste. The icons are scattered across the entire background.

Web Application

Admin Dashboard

Used system configuration, user management, and analytics

The screenshot shows the 'System Overview' page of the Dentify Dental Clinic Management System. The left sidebar contains navigation links: Overview, Clinics, Dentists, Secretaries, Patients, Radiology Centers, Admin Management, and Analytics. The main content area features a 'System Overview' section with four summary cards: Total Clinics (2), Pending Dentists (1), Radiology Centers (2), and Total Patients (5). Below these are two sections: 'Pending Approvals' showing a list of pending approvals for a Secretary and a General Dentistry, and 'Recent Activities' showing a list of recent registrations and admin actions, all marked as 'ACTIVE'.

System Overview
Manage and monitor your dental platform

Total Clinics: 2 | Pending Dentists: 1 | Radiology Centers: 2 | Total Patients: 5

Pending Approvals
Dentists and Secretaries waiting for approval

Test 1
Secretary Jan 15, 2026

Recent Activities
Latest registrations

- New Dentist Registered
Dr. Dentist 4
Jan 20, 2026
ACTIVE
- New Dentist Registered
Dr. Dentist 3
Jan 20, 2026
ACTIVE
- New Admin Added
Admin 1
Jan 15, 2026
ACTIVE
- New Clinic Registered
Clinic 1
Jan 15, 2026
ACTIVE
- New Patient Registered
Ala'a Abdelrahim
Dec 9, 2025
ACTIVE

The screenshot shows the 'Dentist Management' page of the Dentify Dental Clinic Management System. The left sidebar is the same as the previous screenshot. The main content area features a search bar and filters for Dentists. Below is a table listing dentists with columns for Dentist, License, Clinic & Location, Status, Registration Date, and Actions.

Dentist Management
Review and approve dentist registrations

Search dentists by name, email, or license... | All Cities | All Statuses | All Specializations | Clear Filters

DENTIST	LICENSE	CLINIC & LOCATION	STATUS	REGISTRATION DATE	ACTIONS
Dr. Dentist 4 test4dr@gmail.com	123522341	Happy Smile Clinic Jericho	Active	Jan 20, 2026	View Edit Delete
Dr. Dentist 3 test3dr@gmail.com	132114141	Happy Smile Clinic Jaffa	Active	Jan 20, 2026	View Edit Delete
Dr. Dentist 2 test2dr@gmail.com	213131	Clinic 1 Jericho	Pending	Jan 15, 2026	View Edit Approve Reject
Dr. Dentist 1 test1dr@gmail.com	7546366	Happy Smile Clinic Ramallah	Inactive	Jan 15, 2026	View Edit Reactivate
Dr. Ala'a Abdelrahim dentist@gmail.com	DNT56789	Happy Smile Clinic Nabulus	Active	Dec 1, 2025	View Edit Delete

Showing 5 results

Clinic Dashboard

Can manage dentists, secretaries, services, and schedules

The screenshot shows the 'Overview' dashboard for 'Happy Smile Clinic'. The left sidebar contains navigation links: Overview, Appointments, Treatments, Payments, Patients, Dentists, Secretaries, and Analytics. The main content area features a 'Welcome to Happy Smile Clinic' message with the registration number 'Nablus • Registration: CLN12345'. Below this are four summary cards: Secretaries (1), Dentists (4), Today's Appointments (0), and Total Patients (0). A section titled 'Upcoming Appointments' lists three cancelled appointments for 'Abdullah Shabib' on January 26 and 27, 2020, at 02:00 AM, with Dr. Ala'a Abdelrahim as the attending dentist. A 'View All' button is located to the right of the appointments list.

Overview

Welcome to Happy Smile Clinic
Nablus • Registration: CLN12345

Secretaries: 1

Dentists: 4

Today's Appointments: 0

Total Patients: 0

Pending: 0

Upcoming Appointments
Next scheduled appointments

Abdullah Shabib **CANCELLED**
Mon, Jan 26, 02:00 AM
Dr. Ala'a Abdelrahim

Abdullah Shabib **CANCELLED**
Tue, Jan 27, 02:00 AM
Dr. Ala'a Abdelrahim

Abdullah Shabib **CANCELLED**
Tue, Jan 27, 02:00 AM
Dr. Ala'a Abdelrahim

The screenshot shows the 'Dentists Management' page. The left sidebar is identical to the Overview page. The main content area has a title 'Dentists Management' and a subtitle 'Manage dentist requests and profiles for your clinic'. A '+ Request New Dentist' button is in the top right. Below the title is a search bar and two dropdown menus for 'All Specializations' and 'All Status', with a 'Clear Filters' button. A table lists four dentists with columns for Dentist, License & Specialization, Contact, Status, and Actions. All dentists are listed as 'Active'. The table shows details for Dr. Dentist 4, Dr. Dentist 3, Dr. Dentist 1, and Dr. Ala'a Abdelrahim. A 'Showing 4 results' message is at the bottom of the table.

Dentists Management
Manage dentist requests and profiles for your clinic

+ Request New Dentist

Search dentists...

All Specializations All Status Clear Filters

DENTIST	LICENSE & SPECIALIZATION	CONTACT	STATUS	ACTIONS
Dr. Dentist 4 Jawiche	123522341	test4dr@gmail.com +9705981225474	Active	View Edit Delete
Dr. Dentist 3 Jaffa	132114141	test3dr@gmail.com +970598743703	Active	View Edit Delete
Dr. Dentist 1 Ramallah	7546366	test1dr@gmail.com +9705984342421	Active	View Edit Delete
Dr. Ala'a Abdelrahim Nablus	DNT56789	dentist@gmail.com +97059777777	Active	View Edit Delete

Showing 4 results

Dentist Dashboard (1)

Can manage appointments and daily schedule

The screenshot shows the 'Appointments' section of the Dentify Dental Clinic Management System. The interface includes a sidebar with navigation options: Overview, Appointments (selected), Treatments, Payments, Radiology, My Patients, Reports, and Analytics. The main content area is titled 'Appointments' with the subtitle 'Manage and track your daily appointment schedule'. It features filters for 'Today (2)', 'Pending (3)', 'Upcoming (8)', 'Past (20)', and 'All Status (45)'. A search bar and a status dropdown are also present. Below these, a table displays appointment details:

DATE & TIME	PATIENT	TREATMENT	STATUS	ACTIONS
Jan 25, 2026 10:30 AM - 11:00 AM	Abdullah Shabib +97059555555	Orthodontics	Confirmed	[Icons]
Jan 25, 2026 01:45 PM - 02:15 PM	Abdullah Shabib +97059555555	Orthodontics	Confirmed	[Icons]

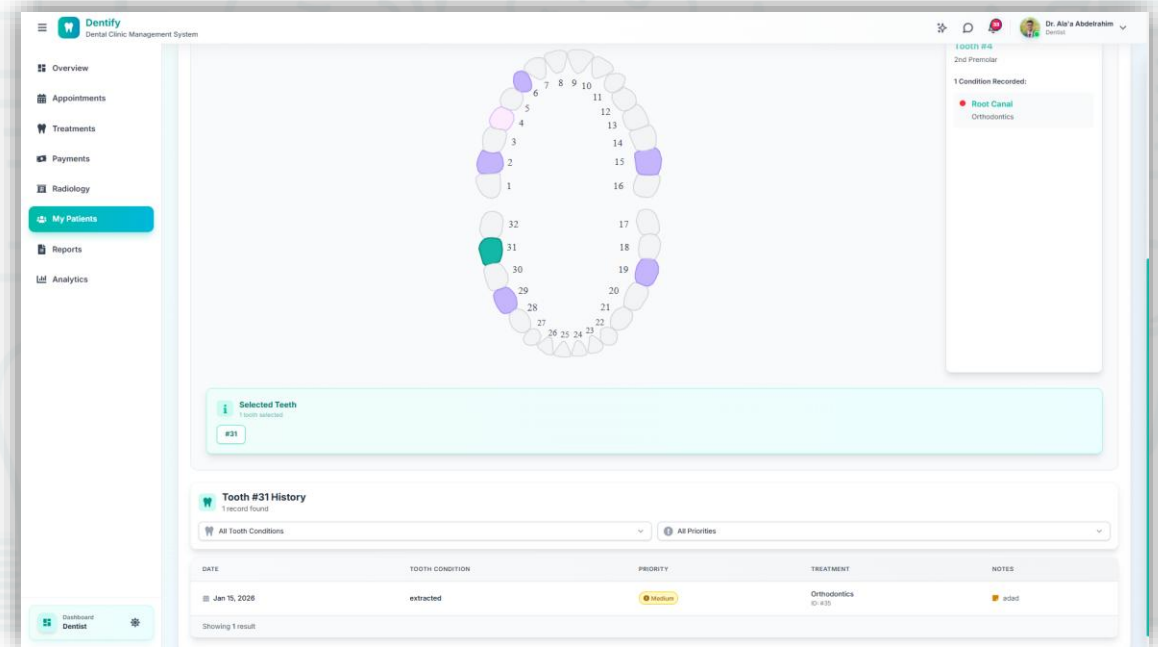
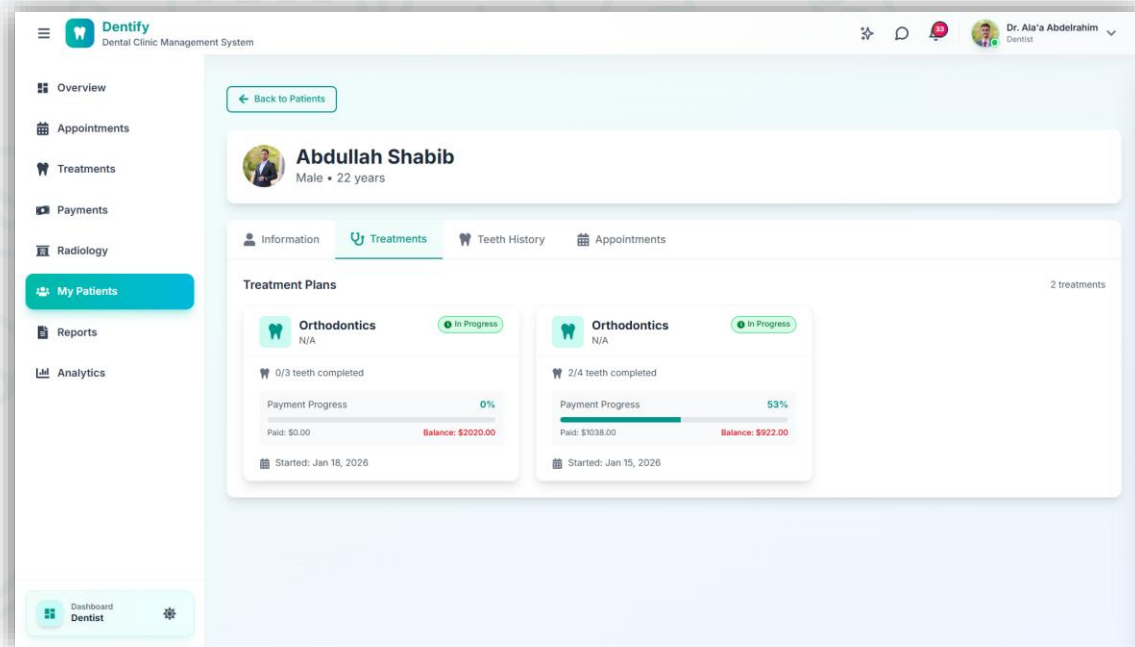
Showing 2 results

The screenshot shows the 'Daily Schedule' view of the Dentify Dental Clinic Management System. The interface includes a sidebar with navigation options: Overview (selected), Appointments, Treatments, Payments, Radiology, My Patients, Reports, and Analytics. The main content area is a calendar grid for the week of January 25 to 31, 2026. The grid shows appointments for Dr. Ala'a Abdelrahim, including Orthodontics and Consultation slots, and break times. The appointments are color-coded: green for confirmed, red for break, and yellow for consultation.

Time	Sun 25	Mon 26	Tue 27	Wed 28	Thu 29	Sat 31
9:00 AM						Closed
9:45 AM						Closed
10:00 AM						Abdullah Shabib Orthodontics 10:00 AM - 10:30 AM
10:30 AM	Abdullah Shabib Orthodontics 10:30 AM - 11:00 AM	Abdullah Shabib Orthodontics 10:30 AM - 11:00 AM				
10:45 AM						
11:15 AM						
11:30 AM						
12:00 PM	Break 12:00 PM - 1:00 PM	Break 12:00 PM - 1:00 PM	Break 12:00 PM - 1:00 PM	Break 12:00 PM - 1:00 PM	Break 12:00 PM - 1:00 PM	Break 12:00 PM - 1:00 PM
1:00 PM						
1:45 PM	Abdullah Shabib Consultation 1:45 PM - 2:15 PM					
2:30 PM			Abdullah Shabib Consultation 2:30 PM - 3:15 PM			Closed
3:15 PM				Abdullah Shabib Consultation 3:15 PM - 4:00 PM		Closed
4:00 PM					Abdullah Shabib Consultation 4:00 PM - 4:45 PM	Closed

Dentist Dashboard (2)

They can also access patient records and treatment history



Dentist Dashboard (3)

Dentists can create treatment plans

Dashboard

Dentist

Overview

Appointments

Treatments

Payments

Radiology

My Patients

Reports

Analytics

Dr. Ala'a Abdelrahim
Dentist

⚙️ 🔔 📞

← Back to Treatments

Create Treatment

New Treatment Plan

Create treatment from appointment

Basic Info

Teeth

Patient Information

Select Patient * (Cannot be changed)

Abdullah Shabib

Patient: Abdullah Shabib
+97059555555

Treatment Details

Treatment Type *

Select an option

Description

Describe the treatment plan and procedures...

[illegible]

Dentist Dashboard (4)

Also create prescriptions

 **Dentify Clinic**
Medical Prescription

Patient Name: Abdullah Shabib
Treatment: Orthodontics
Date: Jan 23, 2026

 **Medications**

1. Accamol
Dosage: 500mg
Duration: 2

Frequency: 23

Doctor's Signature

This prescription is generated electronically and is valid.
For any queries, please contact the clinic.

 **Create Prescription**
For Abdullah Shabib

Patient:
Abdullah Shabib

Treatment:
Orthodontics

Date:
Jan 23, 2026

 **Medications** + Add Medication

Medication 1
Medication Name *
Accamol

Dosage
500mg

Frequency
23

Duration

 Preview & Print

 Cancel

 Save Prescription

Dentist Dashboard (5)

The system allows dentists track payments and financial status

The screenshot shows the 'Payment Management' section of the Dentify Dental Clinic Management System. The interface includes a sidebar with navigation options: Overview, Appointments, Treatments, Payments (selected), Radiology, My Patients, Reports, and Analytics. The main content area has a title 'Payment Management' with a subtitle 'Track and manage treatment payments' and a '+ Record Payment' button. Below this is a search bar and filter controls for 'All Patients', 'All Time', and 'All Methods'. A table displays 5 results of payments, with columns for DATE, PATIENT, TREATMENT, AMOUNT, DISCOUNT, METHOD, NOTES, and ACTIONS. Each row includes an 'Invoice' button. At the bottom, it says 'Showing 5 results'.

DATE	PATIENT	TREATMENT	AMOUNT	DISCOUNT	METHOD	NOTES	ACTIONS
Jan 18, 2026	Patient 1	Orthodontics	\$500.00	\$20.00	Cash	-	Invoice
Jan 17, 2026	Abdullah Shabib	Orthodontics	\$20.00	\$30.00	Cash	adad	Invoice
Jan 15, 2026	Abdullah Shabib	Orthodontics	\$1000.00	\$0.00	Cash	ada	Invoice
Jan 15, 2026	Abdullah Shabib	Orthodontics	\$18.00	\$10.00	Cash	ad	Invoice
Jan 15, 2026	Patient 1	GOOD APPOIT	\$100.00	\$20.00	Cash	bad	Invoice

Showing 5 results

The screenshot shows the 'Orthodontics' section of the Dentify Dental Clinic Management System, specifically the 'Payments' tab. The interface includes a sidebar with navigation options: Overview, Appointments, Treatments (selected), Payments, Radiology, My Patients, Reports, and Analytics. The main content area has a title 'Orthodontics' with a subtitle 'Patient: Abdullah Shabib' and a 'In Progress' status. Below this are tabs for 'Overview', 'Payments' (selected), 'Appointments', and 'Prescriptions'. A summary section shows four key figures: Total Cost (\$2000.00), Total Discount (\$40.00), Total Paid (\$1038.00), and Balance Due (\$922.00). A 'Payment Progress' section shows a progress bar at 53% Complete and 3 Payments. A 'Payment History' section includes a 'Print Full Receipt' button and a table displaying 3 results of payments, with columns for DATE, PATIENT, TREATMENT, AMOUNT, DISCOUNT, METHOD, and NOTES. At the bottom, it says 'Showing 3 results'.

DATE	PATIENT	TREATMENT	AMOUNT	DISCOUNT	METHOD	NOTES
Jan 17, 2026	Abdullah Shabib	Orthodontics	\$20.00	\$30.00	Cash	adad
Jan 15, 2026	Abdullah Shabib	Orthodontics	\$1000.00	\$0.00	Cash	ada
Jan 15, 2026	Abdullah Shabib	Orthodontics	\$18.00	\$10.00	Cash	ad

Showing 3 results

Dentist Dashboard (6)

Send and receive radiology requests

The screenshot shows the 'Radiology Requests' section of the Dentify Dental Clinic Management System. The interface includes a sidebar with navigation options: Overview, Appointments, Treatments, Payments, Radiology (highlighted), My Patients, Reports, and Analytics. The main content area is titled 'Radiology Requests' with a subtitle 'Manage diagnostic imaging requests' and a '+ New Request' button. Below this is a search bar and filter options for 'All Status' and 'All Types'. A table displays a list of requests with columns for Patient, Imaging Type, Radiology Center, Request Date, Available Date, Status, and Actions. The table shows 7 results, with some requests marked as 'Requested' or 'Completed'.

PATIENT	IMAGING TYPE	RADIOLOGY CENTER	REQUEST DATE	AVAILABLE DATE	STATUS	ACTIONS
Abdullah Shabib	X-Ray	Happy Smile Center	Jan 22, 2026	Pending	Requested	[Icon]
Abdullah Shabib	X-Ray	Happy Smile Center	Jan 18, 2026	Jan 18, 2026	Completed	[Icon] [Icon]
Patient 1	X-Ray	Happy Smile Center	Jan 17, 2026	Pending	Requested	[Icon]
Abdullah Shabib	X-Ray	Center 1	Jan 17, 2026	Pending	Requested	[Icon]
Abdullah Shabib	CT	Happy Smile Center	Jan 15, 2026	Pending	Requested	[Icon]
Patient 1	CT	Happy Smile Center	Jan 15, 2026	Jan 15, 2026	Completed	[Icon]
Abdullah Shabib	X-Ray	Happy Smile Center	Jan 15, 2026	Jan 22, 2026	Completed	[Icon] [Icon]

Showing 7 results

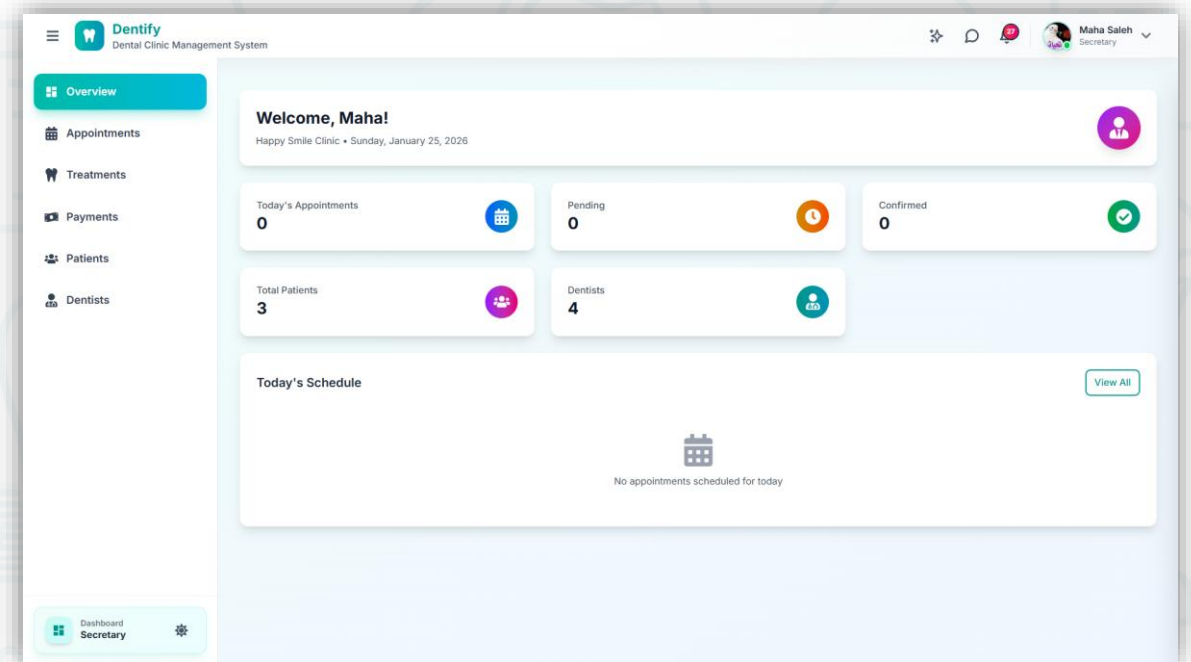
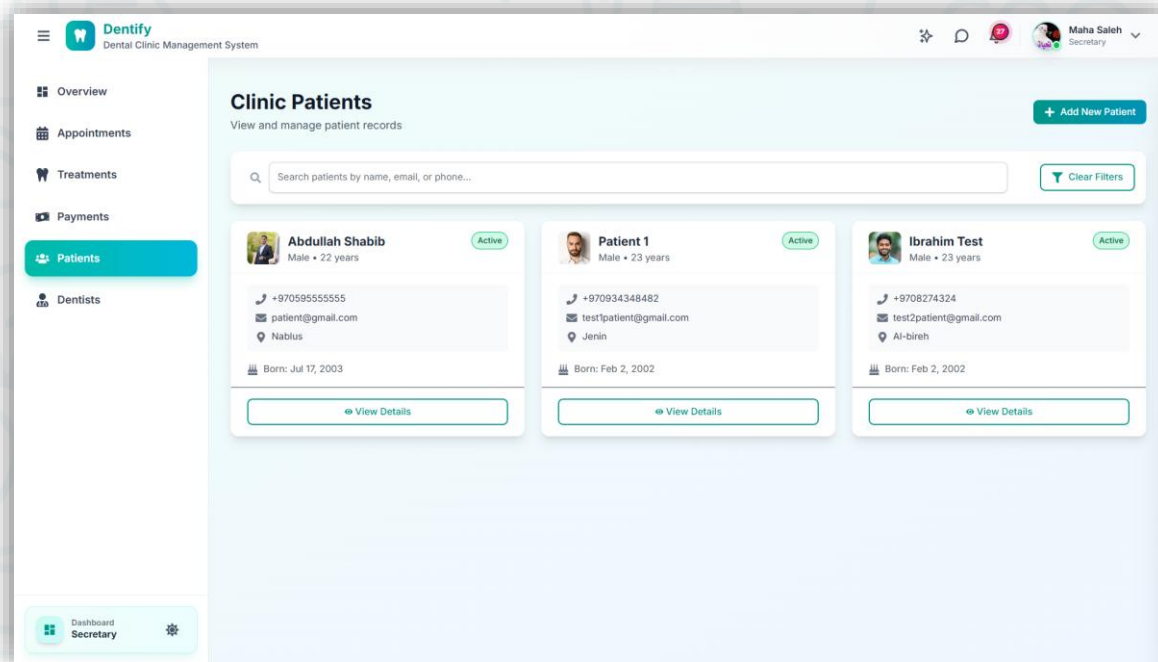
The screenshot shows the 'New Radiology Request' form, which is used to request diagnostic imaging. The form includes the following fields:

- Patient ***: A dropdown menu to select a patient.
- Radiology Center ***: A dropdown menu to select a radiology center.
- Related Treatment (Optional)**: A dropdown menu to select a patient first, with a note to select a patient to see their treatments.
- Imaging Type ***: A dropdown menu to select an imaging type, with a note to select a radiology center to see available types.
- Notes (Optional)**: A text area to add any additional notes or special instructions.

At the bottom of the form are two buttons: 'Cancel' and 'Create Request'.

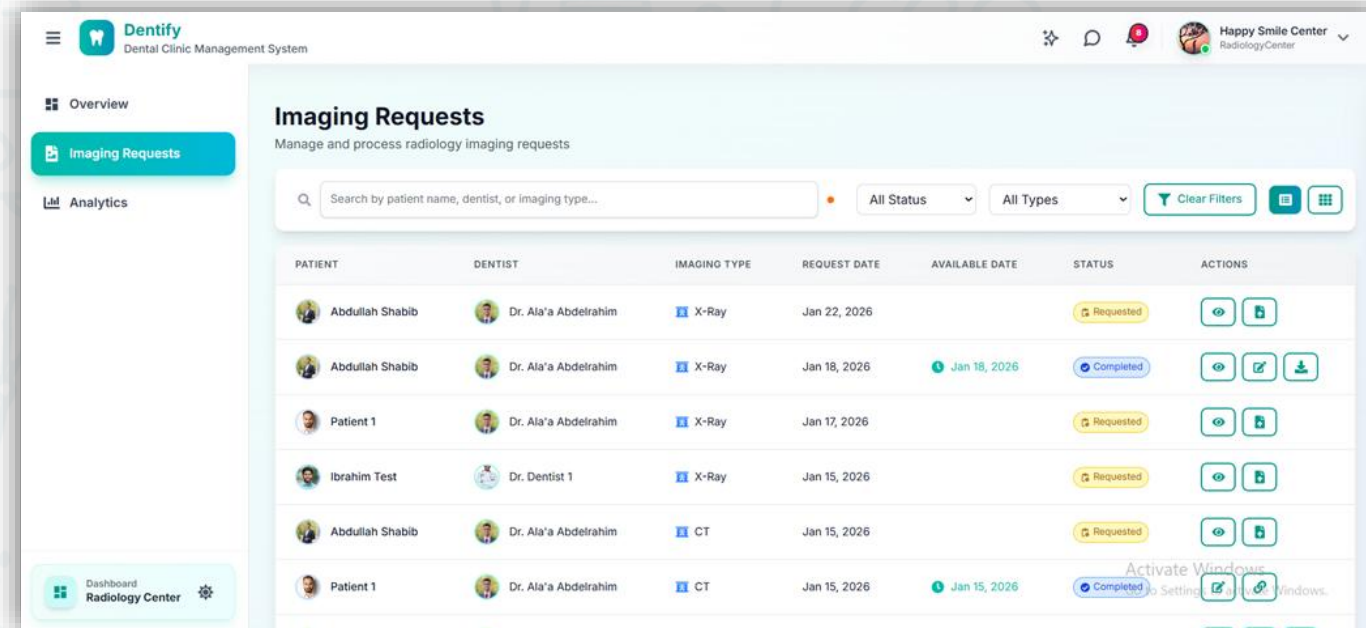
Secretary Dashboard

Handle daily scheduling, appointment management, and patient follow-up



Radiology Center Dashboard

Receive imaging requests and upload medical results



Dentify
Dental Clinic Management System

Happy Smile Center
RadiologyCenter

Overview
Imaging Requests
Analytics

Imaging Requests

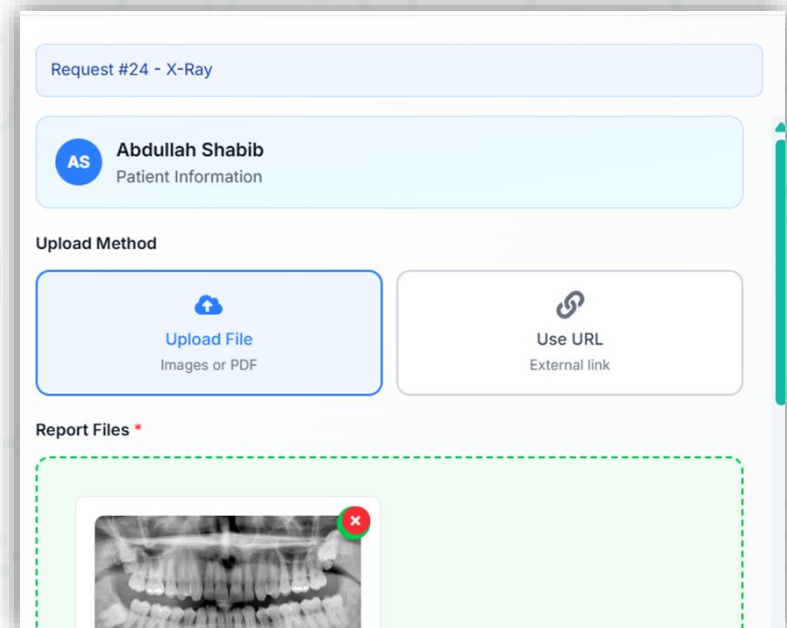
Manage and process radiology imaging requests

Search by patient name, dentist, or imaging type...

All Status All Types Clear Filters

PATIENT	DENTIST	IMAGING TYPE	REQUEST DATE	AVAILABLE DATE	STATUS	ACTIONS
Abdullah Shabib	Dr. Ala'a Abdelrahim	X-Ray	Jan 22, 2026		Requested	 
Abdullah Shabib	Dr. Ala'a Abdelrahim	X-Ray	Jan 18, 2026	Jan 18, 2026	Completed	  
Patient 1	Dr. Ala'a Abdelrahim	X-Ray	Jan 17, 2026		Requested	 
Ibrahim Test	Dr. Dentist 1	X-Ray	Jan 15, 2026		Requested	 
Abdullah Shabib	Dr. Ala'a Abdelrahim	CT	Jan 15, 2026		Requested	 
Patient 1	Dr. Ala'a Abdelrahim	CT	Jan 15, 2026	Jan 15, 2026	Completed	  

Dashboard Radiology Center



Request #24 - X-Ray

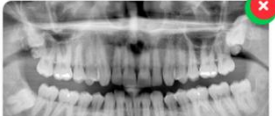
Abdullah Shabib
Patient Information

Upload Method

 **Upload File**
Images or PDF

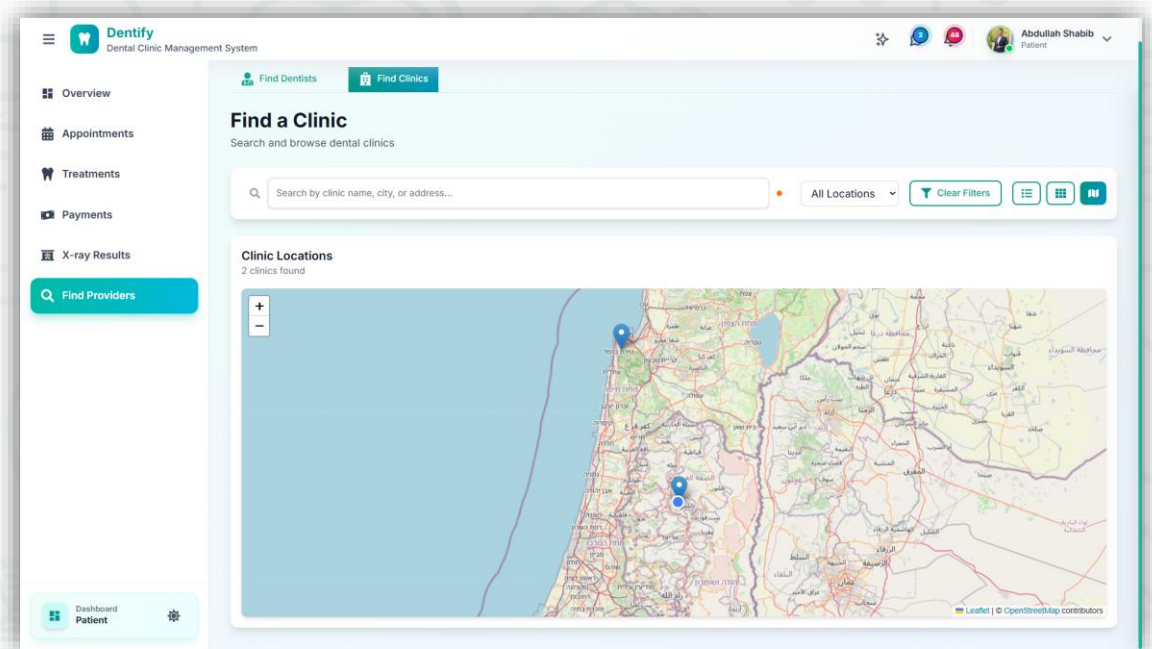
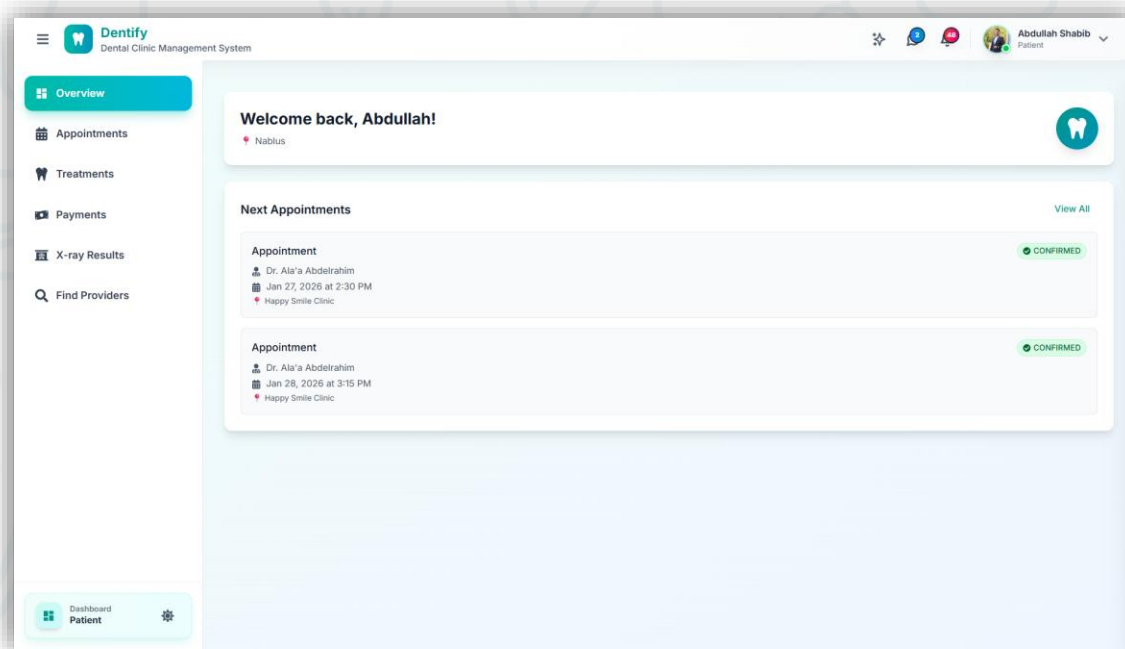
 **Use URL**
External link

Report Files *



Patient Dashboard

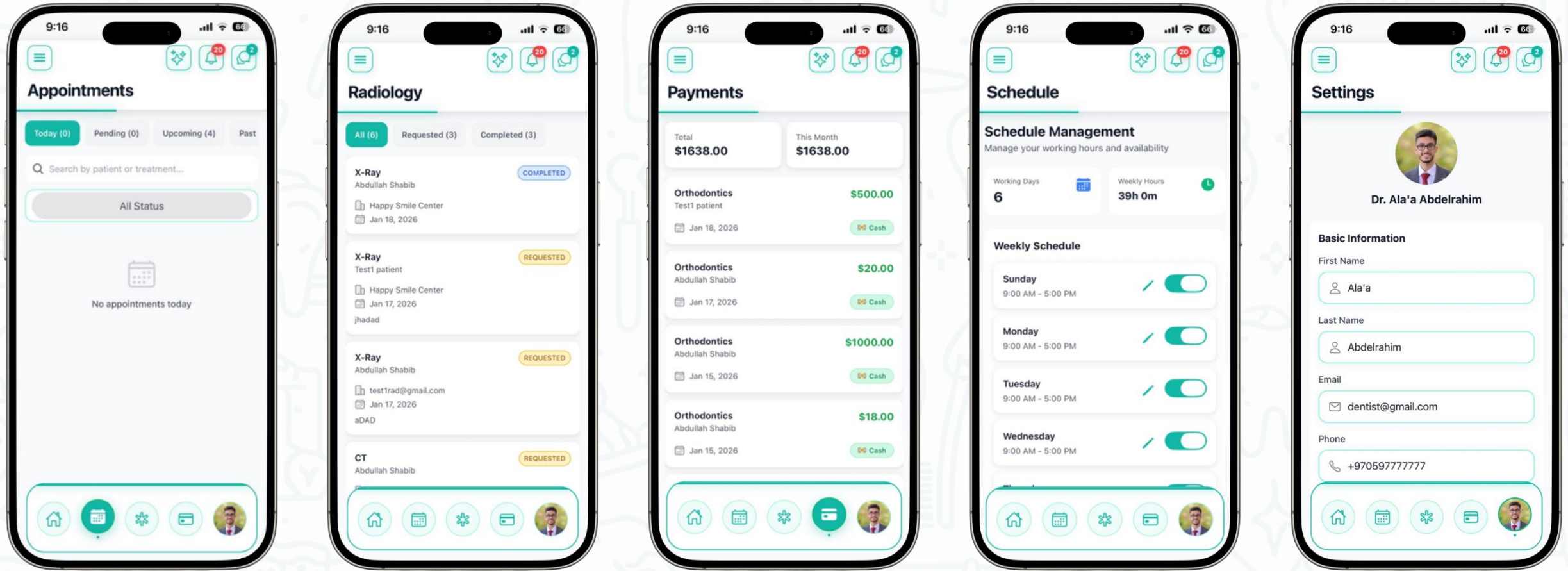
Can book appointments, view records, and payments



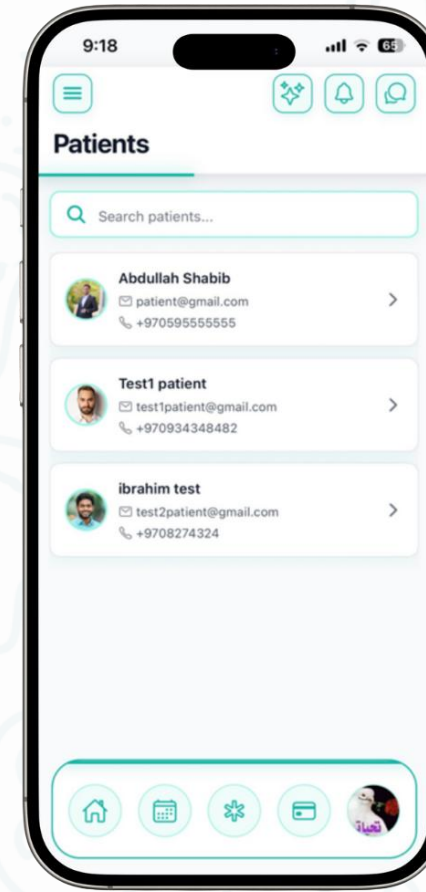
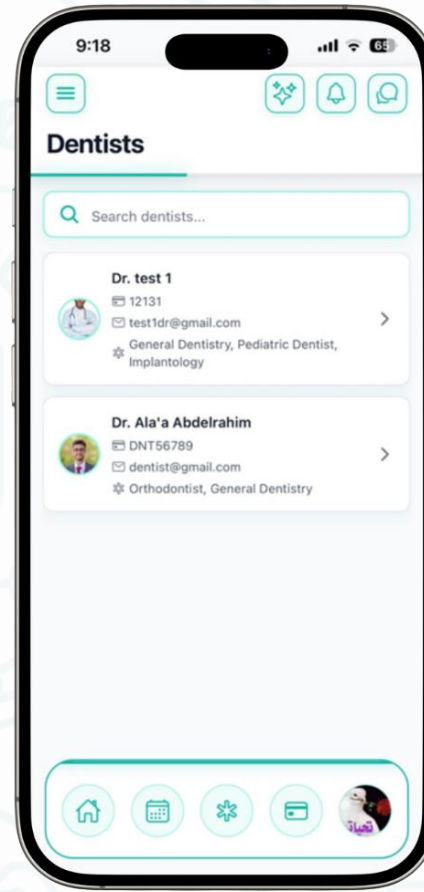
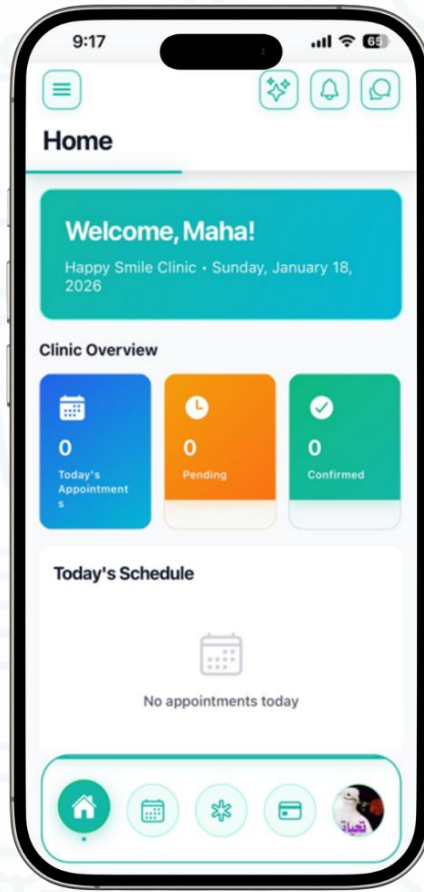
The background is a light gray pattern of various dental-related icons. These include teeth, toothbrushes, dental X-rays, dental chairs, dental tools like pliers and mirrors, and bottles of mouthwash or toothpaste. The icons are scattered across the entire background.

Mobile App

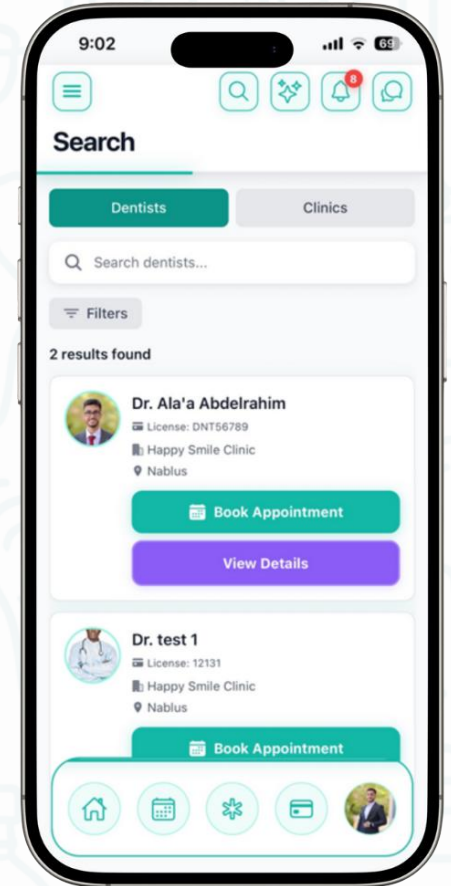
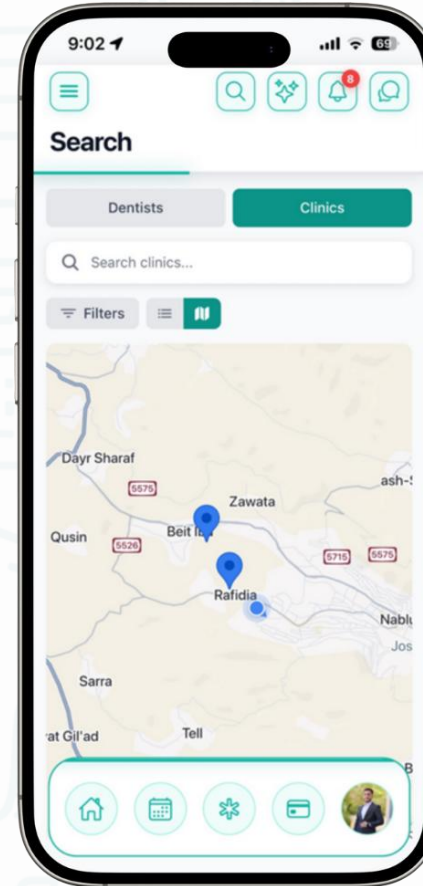
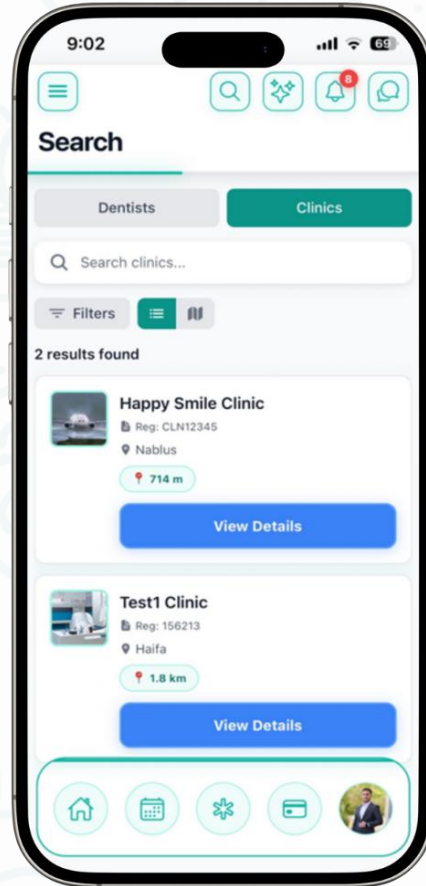
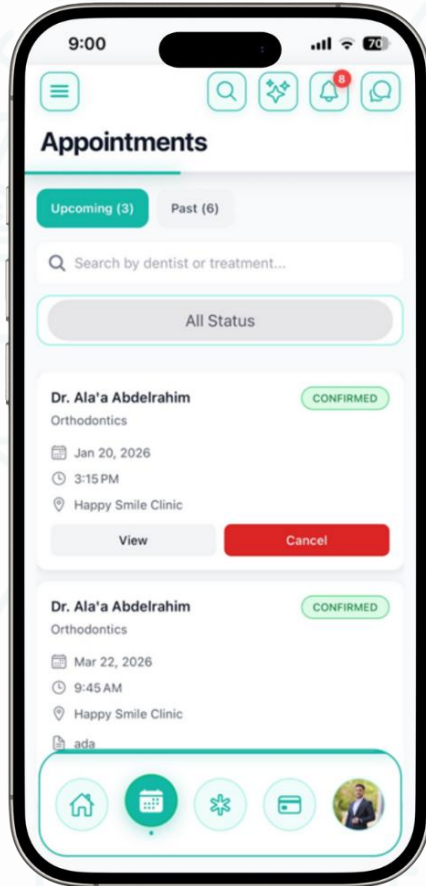
Dentist Dashboard



Secretary Dashboard



Patient Dashboard



Results

The system successfully:

- Centralized clinic operations
- Reduced administrative workload
- Improved patient experience
- Enabled AI-driven automation
- Worked across web and mobile platforms

Future Work

- **Rating & feedback system**

Patients rate dentists and clinic services

- **Enhanced AI features**

Smarter recommendations and expanded chatbot functions

- **Tele-dentistry**

Online consultations and follow-ups.

- **Multi-language support**

Improve accessibility for wider populations.

Conclusion

- **Modern & efficient**

Dentify modernizes dental clinic management.

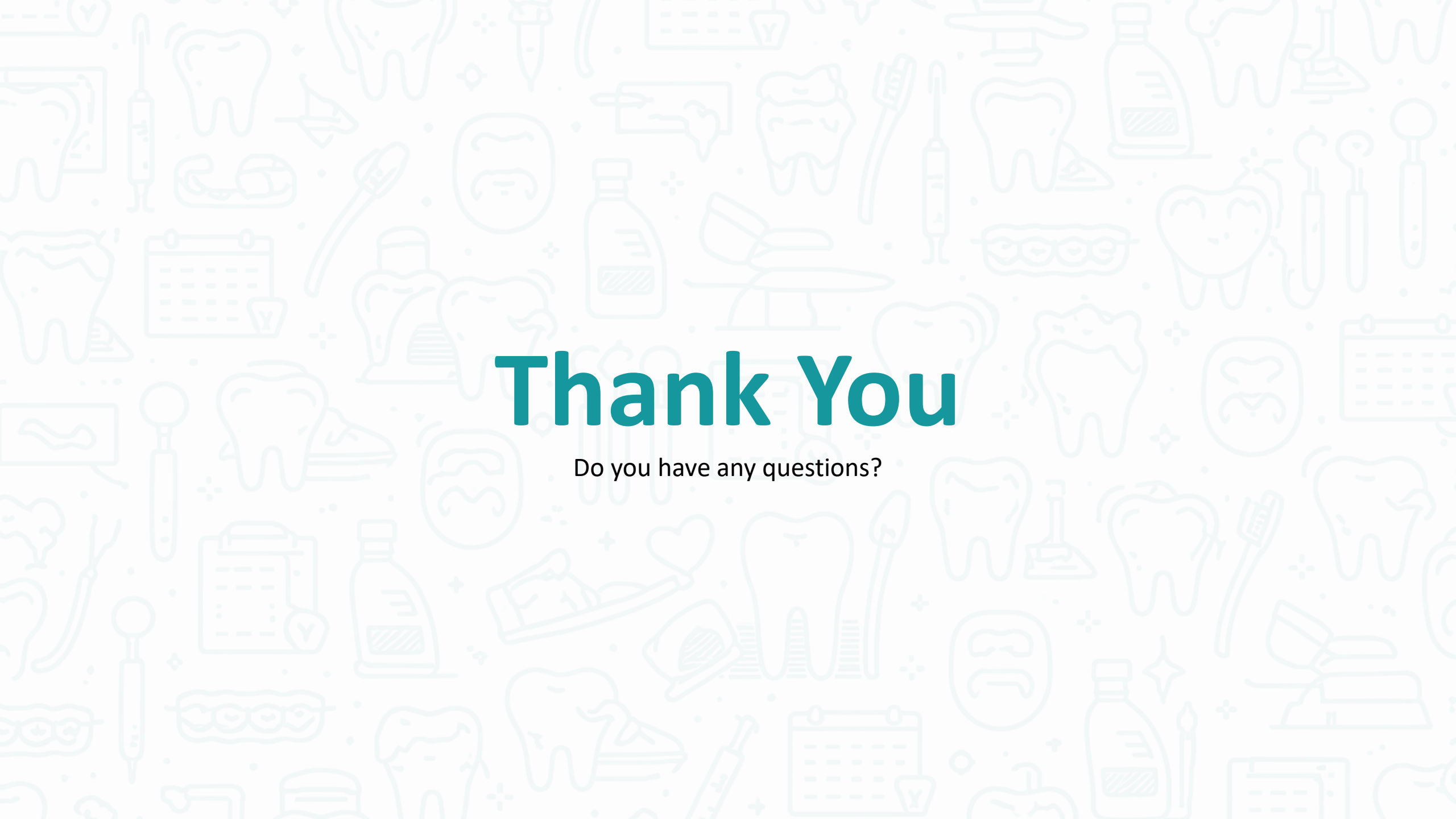
- **Combines AI & healthcare**

Brings intelligent automation into daily practice.

- **Improves accessibility**

For clinics, dentists, and patients alike.

Dentify demonstrates how software engineering can directly improve healthcare services.

The background is a light gray pattern of various dental-related icons. These include teeth, toothbrushes, dental X-rays, dental chairs, dental tools like pliers and mirrors, and bottles of mouthwash or toothpaste. The icons are scattered across the entire background.

Thank You

Do you have any questions?